EPIDEMIOLOGICAL DATA ANALYSIS USING MICROSOFT EXCEL

Analysis of Malaria Cases: A Public Health Data Analysis

Project Type: Public Health / Epidemiology / Data Analysis

Tools Used: WPS Spreadsheet

Dataset: 100 synthetic patient records

1. Introduction

Malaria is an important public health concern in many parts of the world. Epidemiological data analysis helps organize health information, identify patterns and communicate findings that can inform further investigation.

This project demonstrates the use of spreadsheet tools to clean, summarize, and visualize a synthetic malaria dataset.

2. Objectives

The objectives were to:

- Clean and organize a dataset containing 100 fictional patient records.
- Summarize positive and negative malaria test results.
- Compare positivity rates by sex, age group, and community.
- Create charts to communicate the findings clearly.
- Demonstrate basic epidemiological data-analysis skills using a spreadsheet.

3. Methodology

A synthetic dataset of 100 fictional patient records was used for this practice project. Variables included patient ID, age, sex, community, month, malaria test result, fever status, treatment, and outcome.

Data cleaning involved standardizing inconsistent entries and identifying missing age values. Missing ages were left blank rather than estimated.

PivotTables and spreadsheet formulas were used to calculate frequencies and positivity rates. Charts were created to present selected results.

The positivity rate was calculated as:

Positivity rate = Positive test results ÷ Total test results × 100

4. Results

4.1 Overall malaria test results

Of the 100 synthetic records, 52 had positive malaria test results and 48 had negative results. The overall positivity rate was 52%.

4.2 Results by sex

- Female: 29 positive results out of 60 records, giving a positivity rate of 48.3%.
- Male: 23 positive results out of 40 records, giving a positivity rate of 57.5%.

The synthetic dataset showed a higher positivity rate among males than females.

4.3 Results by age group

Among the 97 records with recorded ages:

- Ages 0–14: 14 positive out of 22 records (63.6%).
- Ages 15–24: 11 positive out of 16 records (68.8%).
- Ages 25–44: 15 positive out of 35 records (42.9%).
- Ages 45 and above: 9 positive out of 24 records (37.5%).

Three records had missing ages and were excluded from the age-group comparison. The highest positivity rate among the age groups with known ages was observed in the 15–24 group.

4.4 Results by community

- Community A: 14 positive out of 24 records (58.3%).
- Community B: 14 positive out of 24 records (58.3%).
- Community C: 12 positive out of 29 records (41.4%).
- Community D: 12 positive out of 23 records (52.2%).

Communities A and B had the highest positivity rates in this synthetic dataset while Community C had the lowest.

5. Discussion

The analysis demonstrates how spreadsheet-based epidemiological methods can summarize test results and compare patterns across demographic groups and communities.

The differences observed between groups are descriptive findings from a synthetic dataset. They do not establish the causes of the differences or demonstrate statistically significant associations.

6. Limitations

This project uses fictional data created for learning and portfolio demonstration. It does not represent actual patients, communities or malaria prevalence.

The dataset is small and may not reflect real-world population characteristics. Missing ages also limited the age-group analysis.

7. Conclusion

This project demonstrates basic skills in data cleaning, descriptive epidemiological analysis, PivotTables, percentage calculations, and data visualization using spreadsheet software.

These techniques can support the organization and presentation of public health data when applied appropriately to real quality-controlled datasets.

8. Ethical Note

All records in this project are synthetic. No real patient information was used. The results must not be interpreted as actual clinical or public health findings.